



Odoo Cafe POS

Project Overview

Build a complete web-based Restaurant Point-of-Sale system. The system has two parts: a Backend where the user (admin) configures products, employees, tables, payment methods, and promotions, and a POS Terminal which the employee uses to take orders, process payments, and manage the floor. The Kitchen Display runs as a separate screen that updates in real time.

There are three distinct roles in this system:

- **User** – The admin who logs into the backend to configure and manage everything.
- **Employee** – The cashier who operates the POS terminal to take orders and handle payments.
- **Customer** – A person who visits the cafe and is managed from within the POS terminal by the employee.

1. Application Flow

- User signs up or logs in. On successful login, the POS session opens directly.
- Employee sees the Floor pop-up, selects a table, and is taken to the Order View.
- Employee picks products, adjusts quantities, applies discounts or coupons, and sends the order to the Kitchen Display.
- Kitchen staff moves the order through To Cook, Preparing, and Completed stages.
- Employee processes payment via Cash, Card/Digital, or UPI QR. The receipt can be printed or sent via email.
- At the end of the shift, the session is closed and the user reviews reports.

2. Backend – Features to Build

The backend is accessed by the user (admin) to configure everything before and after POS sessions. The navigation includes Products, Category, Payment Method, Coupon & Promotion, Booking, User/Employee, KDS, Reports and Log-Out.

2.1 Login & Signup

Signup form fields: Name, Email, Password.

Login form fields: Email, Password.

On successful login, the POS session opens directly.

2.2 Product Management

User can create, list, update, and delete products.

Product Fields

- Name
- Category
- Price
- Unit of Measure (e.g. per piece, per kg, per litre)
- Tax
- Description

The Category field allows picking an existing category or creating a new one on the fly without leaving the product form.

2.3 Product Category Management

Full create, list, edit, and delete support.

Category Fields

- Name
- Color

The color assigned to a category appears everywhere it is used (e.g. product cards in the POS terminal, category filter tabs, and the order view). If the color is updated in the backend, it reflects across all those places automatically.

2.4 Payment Method Setup

User enables or disables each payment method using a toggle.

Supported Methods:

- **Cash** — Available at checkout when enabled.
- **Digital/Card** — Represents card and bank payments.
- **UPI QR** — Requires a UPI ID to be saved (e.g. cafe@ybl). The system generates a QR code from it dynamically at the payment screen.

2.5 Floor Plan & Table Management

User creates floors and adds tables under each floor.

Table Fields

- Table Number
- Number of Seats
- Active Status

These tables appear in the Floor pop-up inside the POS terminal.

2.6 Coupons & Promotions

Two types of discounts can be created.

Coupon Codes

User creates a coupon with a code and a discount type either a percentage or a fixed amount off the whole order.

The employee enters the code manually in the POS to redeem it.

Automated Promotions

These trigger automatically without any code entry.

If applied to a Product, a Minimum Quantity is set. The discount fires when the employee reaches that quantity.

If applied to an Order, a Minimum Order Amount is set. The discount fires when the cart total crosses that amount.

In both cases, the discount is either a percentage or a fixed amount and applies to the whole order.

2.7 User / Employee Management

User can list all accounts and add new ones.

Fields

- Name
- Email
- Role (User/Admin or Employee/Cashier)

Actions Available on Each Record

- Change Password
- Archive
- Delete

Archive deactivates the account without deleting it.

2.8 POS Terminal & Session

The POS terminal shows the last open session date and last closing sale amount.

The Open Session button launches the terminal.

When the session is closed at the end of the shift, a closing summary is displayed.

2.9 Customer Facing Display

The system provides a Customer Facing Display that allows customers to view order and payment information during checkout.

The Customer Facing Display is accessible through a dedicated URL by appending `/customer-display` to the database URL.

The display updates in real time based on actions performed in the POS terminal.

Order View

The display shows:

- Ordered products
- Quantities
- Prices
- Subtotal
- Tax
- Discounts
- Total amount

Payment View

When the employee proceeds to payment, the display shows the payment information.

For UPI payments, a QR code is displayed along with the payable amount.

Order Completion View

After successful payment, the display shows a confirmation and thank you message.

2.10 Self Ordering

The system supports customer self-ordering through QR codes.

Configuration

The user can enable Self Ordering and choose one of the following modes:

- **Online Ordering** — Customers can browse products and place orders.
- **QR Menu** — Customers can browse the digital menu only.

The user can also configure:

- Background Color
- Background Images

QR Codes

The system automatically generates a unique QR code for each table.

Each QR code contains a URL in the format:

`<domain>/s/<unique-token>`

The unique token identifies the table automatically when a customer scans the QR code.

QR codes can be downloaded as a PDF for printing and placing on tables.

Customer Experience

Customers can:

- Browse products
- Add items to cart
- Apply coupon codes
- Place orders
- Track order status (To Cook, Preparing, Completed)

Orders created through Self Ordering are automatically sent to the Kitchen Display System (KDS).

2.11 Reporting & Dashboard

Provides real-time sales insights. All stats, charts, and tables update automatically when a filter is changed.

Available Filters

- Period (Today, This Week, This Month, or a custom date range)
- Employee
- Session

- Product

Summary Metrics Shown on the Dashboard

- Total Orders
- Revenue
- Average Order Value

Dashboard Components

- Sales Trend Chart showing revenue or order count over time
- Top Categories Chart showing sales distribution by category
- Top Orders Table showing the highest-value orders
- Top Products Table showing product name, quantity sold, and revenue
- Top Categories Table showing category-wise revenue

Reports can be exported as PDF or XLS.

3. POS Terminal – Views to Build

The POS Terminal opens directly after login and is used by the employee to manage orders, customers, payments, and tables.

3.1 Navigation

The top navigation bar contains:

- POS Order — Opens the main order-taking screen.
- Orders — Opens the list of all orders for the current session.
- Customer — Opens customer management.
- Table View — Opens the floor and table selection screen.
- Product Search Bar — Searches products by name.
- Current Table Indicator.
- Employee Icon — Shows the currently logged-in employee.
- Hamburger Menu — Opens a dropdown with links to Products, Category, Payment Method, Coupon & Promotion, Booking, User/Employee, KDS, Reports and Log-Out.

3.2 Floor Pop-up

The Floor Pop-up appears when the session starts or when the employee taps Table View.

It shows all floors with their tables as a numbered grid of clickable cards. Each card displays the table number and number of seats. Tables with active orders are visually distinct from available ones.

Selecting a table opens the Order View for that table.

3.3 Order View

The Order View is the primary screen used by the employee and is divided into three sections: Product, Cart, and Payment.

Product Section

Products are displayed as cards. Category tabs allow the employee to filter products by category. The employee can also search by product name.

Clicking a product adds it to the cart.

Cart Section

Each item shows:

- Product Name
- Quantity
- Unit Price
- Line Total

Quantity can be adjusted directly from the cart.

If a product-level promotion is applied, the discount is shown on the corresponding product line.

If an order-level discount or coupon is applied, it appears as a separate line in the order summary.

A **Send to Kitchen** action allows the employee to send the current order to the Kitchen Display.

Order Summary

- Subtotal
- Tax
- Discounts
- Total

Actions

- **Customer** — Assign a customer to the current order.
- **Discount** — Open the coupon code popup.
- **Send** — Open the receipt email popup.

Payment Section

Shows all payment methods enabled in the backend.

The employee selects a method and completes the transaction.

3.4 Discount Popup

The employee enters a coupon code in the popup. If the code is valid, the discount is applied and reflected in the order summary.

Automated promotions apply automatically and do not use this popup.

3.5 Payment & Receipt

Supported Payment Methods

- **Cash** — Employee enters the amount received and the system shows the change due.
- **UPI** — A QR code is generated from the saved UPI ID and shown on screen with the total amount. The employee clicks Confirmed once the customer has paid or Cancel to go back.
- **Card** — Employee enters a transaction reference.

After successful payment, the order is marked as paid.

The employee can:

- Print the receipt.
- Send the receipt to the customer via email.

3.6 Orders

Shows all orders created during the current session. The search bar filters by customer name, order number, or date.

Each Order Shows

- Order Number
- Date
- Customer
- Amount
- Status (Draft, Paid, or Cancelled)

Clicking an order opens the Order Detail view showing:

- Order Number
- Date
- Customer

- Amount
- Status
- Products

For Draft orders, both the Delete and Edit Order buttons are visible.

Clicking Edit Order redirects the employee back to the cart with that order loaded for editing.

Paid orders are view-only.

3.7 Table View

Shows all tables across all floors. Tables with active orders are visually distinct from available ones. Selecting a table opens the corresponding order.

3.8 Customer Management

The employee can search existing customers or create new ones directly from the POS terminal.

Customer Fields

- Name
- Email
- Phone Number

Actions

- Create
- Edit
- Delete

Once a customer is selected, they are linked to the current order and their email is used for receipt delivery.

4. Kitchen Display

The Kitchen Display is accessed through a fixed system URL and is intended to be opened on a separate device or browser tab by the kitchen staff.

It receives orders in real time when the employee clicks **Send to Kitchen**.

Each order appears as a ticket card showing:

- Order Number (same as the ticket number)
- Ordered Items

- Quantities

Only products assigned to the Kitchen Display appear on this screen.

Order Stages

- To Cook — Order just received, not yet started.
- Preparing — Order currently being made.
- Completed — Order ready to serve.

Clicking a ticket card moves the entire order to the next stage.

Clicking an individual item within a ticket marks only that item as completed with a strikethrough, allowing kitchen staff to track progress item by item.

The Kitchen Display also provides a search bar and filters by product and category.

5. What Participants Will Learn

By completing this challenge, participants will gain hands-on experience with:

- Building a complete full-stack web application
- Designing and consuming REST APIs
- Authentication and role-based access control
- Database design and data relationships
- Real-time order and kitchen workflows
- Payment processing and receipt generation
- Reporting and analytics dashboards
- Managing products, customers, employees, and orders

Participants will also gain insight into how a real restaurant POS system operates, from order creation and kitchen preparation to payment and reporting.

Mockup Link -

<https://link.excalidraw.com//65VNwvy7c4X/1Vvr9oy6B3F>